
Fred Peng

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Education

- 2024– **Ph.D. in Computational Biology and Bioinformatics**, Duke University.
(Expected May 2027)
Advisors: Anru R. Zhang, Alexander Tong, and Christian Dallago.
- 2023 Ph.D. student in Computer Science, University of Missouri.
- 2019–2023 **B.E. in Computer Science**, East China Jiaotong University. GPA: 3.72/4.0.

Publications

* Denotes equal contribution. † Denotes corresponding author.

Google Scholar: scholar.google.com/citations?user=bZwJ9oUAAAAJ

- [6] **Peng, F. Z.**, Bezemek, Z., Rector-Brooks, J., Zhang, S., Zhang, A. R., Bronstein, M., Bose, A. J.[†] & Tong, A.[†] *Planner Aware Path Learning in Diffusion Language Models Training*. International Conference on Learning Representations (ICLR) **Oral** (top 1.2%) (2026).
- [5] Wang, C., Zhang, Y., Fang, M., **Peng, Z.** & Cao, L. *Frontiers and Challenges in the Design of Binders for Intrinsically Disordered Proteins*. Current Opinion in Structural Biology 98, 103256 (2026).
- [4] **Peng, F. Z.**, Wang, C., Chen, T., Schussheim, B., Vincoff, S. & Chatterjee, P.[†] *PTM-Mamba: A PTM-Aware Protein Language Model with Bidirectional Gated Mamba Blocks*. Nature Methods 22, 5475–5478 (2025).
- [3] Rector-Brooks, J., Hasan, M., **Peng, Z.**, Quinn, Z., Liu, C., Mittal, S., Dziri, N., Bronstein, M., Bengio, Y., Chatterjee, P., Tong, A.[†] & Bose, A. J.[†] *Steering Masked Discrete Diffusion Models via Discrete Denoising Posterior Prediction*. International Conference on Learning Representations (ICLR) (2025).
- [2] Jing, H., Gao, Z., Xu, S., Shen, T., **Peng, Z.**, He, S., You, T., Ye, S., Lin, W. & Sun, S. *Accurate Prediction of Antibody Function and Structure Using Bio-Inspired Antibody Language Model*. Briefings in Bioinformatics 25(4), bbae245 (2024).
- [1] Wang, C., Qu, Y., **Peng, Z.**, Wang, Y., Zhu, H., Chen, D. & Cao, L. *Proteus: Exploring Protein Structure Generation for Enhanced Designability and Efficiency*. International Conference on Machine Learning (ICML) (2024).

Peer-Reviewed Workshop Papers

- [4] **Peng, F. Z.**^{*}, Bezemek, Z.^{*}, Patel, S., Rector-Brooks, J., Yao, S., Bose, A. J., Tong, A.[†] & Chatterjee, P.[†] *Path Planning for Masked Diffusion Model Sampling*. DeLTa Workshop at ICLR (2025). **Outstanding Paper Award**.
- [3] Patel, S., **Peng, F. Z.**, Fraser, K., Friedman, A. D., Chatterjee, P.[†] & Yao, S. *EvoFlow-RNA: Generating and Representing Non-Coding RNA with a Language Model*. ICLR Workshop on AI for Nucleic Acids (2025).
- [2] Patel, S., Fraser, K., **Peng, Z.**, Friedman, A. D., Yao, O., Chatterjee, P.[†] & Yao, S. *AptaBLE: A Deep Learning Platform for SELEX Optimization*. NeurIPS Workshop on AI for New Drug Modalities (2024).
- [1] **Peng, Z.** & Huang, X. *Spatial-Temporal Transformer Network with Self-Supervised Learning for Traffic Flow Prediction*. IJCAI-ECAI Workshop on Spatial-Temporal Reasoning and Learning (2022).

Preprints

- [10] **Peng, F. Z.**, Zheng, K. & Zhang, A. R. *Distilled Continuous Diffusion Language Models Can Write Code in Few Steps—or One*. arXiv:2609.04531 (2026).
- [9] Lyu, P., Henao, R., Kwee, L. C., **Peng, F. Z.**, Hurdle, M., Vemulapalli, S., Shah, S. H., Khouri, M. G. & Zhang, A. *Synthetic Echocardiograms from Diffusion Models in Rare Cardiovascular Disease*. medRxiv (2026).
- [8] Zhang, S., Liu, X., **Peng, F. Z.**, Yang, Z., Wu, S., Ma, Y. & Zhang, J. *Off-Manifold Collapse in Guided Protein Language Models*. arXiv:2608.18597 (2026).
- [7] Li, M., Chen, Q., Seo, J., Liu, Y., **Peng, F. Z.**, Huang, K., Dai, Z., Chen, C., Zhang, Z., Zhao, J., Wang, Z., Yuan, F., Ma, X., Martinez, D. R. & Li, D. *Epitope-Focused Immunogens Confer Broad Protection against Coronaviruses*. bioRxiv (2026).
- [6] **Peng, F. Z.**, Bose, A. J., Zhang, A. R. & Tong, A. *Coupling Models for One-Step Discrete Generation*. arXiv:2605.07193 (2026).
- [5] **Peng, F. Z.**, Fox, A., Zhang, A. R. & Tong, A. *Don't Retrain, Align: Adapting Autoregressive LMs to Diffusion LMs via Representation Alignment*. arXiv:2605.06885 (2026).
- [4] Zhang, S., Zhuang, C., Cui, C., Yang, Z., **Peng, F. Z.**, Zhang, Y., Bai, H., Jia, Z., Zhou, Y., Chen, G. & Liu, M. *Expert-Choice Routing Enables Adaptive Computation in Diffusion Language Models*. arXiv:2604.01622 (2026).
- [3] Zhang, S., **Peng, F. Z.**, Zhang, Y., Pan, J. & Chrysos, G. G. *Corrective Diffusion Language Models*. arXiv:2512.15596 (2025).
- [2] **Peng, Z.**, Han, C., Wang, X., Li, D. & Yuan, F. *Generative Diffusion Models for Antibody Design, Docking, and Optimization*. bioRxiv (2023).
- [1] Shen, T.^{*}, Hu, Z.^{*}, **Peng, Z.**^{*}, Chen, J., Xiong, P., Hong, L., Zheng, L., Wang, Y., King, I., Wang, S., Sun, S. & Li, Y. *E2EFold-3D: End-to-End Deep Learning Method for Accurate de Novo RNA 3D Structure Prediction*. arXiv:2207.01586 (2022).

Open-Source Projects

- **Open-dLLM** — Open diffusion language models for code generation: pretraining, evaluation, inference, and open checkpoints.
github.com/pengzhangzhi/Open-dLLM 656 stars
- **FAPLM** — A drop-in efficient PyTorch implementation of protein language models.
github.com/pengzhangzhi/faplm 168 stars
- **Awesome-Mamba** — Curated list of papers that extend Mamba to various applications.
github.com/pengzhangzhi/Awesome-Mamba 142 stars
- **self-taught-CS** — Notes and lab code for public computer science courses.
github.com/pengzhangzhi/self-taught-CS 133 stars
- **ab_opt** — Official implementation of *Generative Diffusion Models for Antibody Design, Docking, and Optimization*.
github.com/pengzhangzhi/ab_opt 63 stars
- **Awesome-Computational-Structural-Biology** — Curated self-learning materials in computational structural biology.
github.com/pengzhangzhi/Awesome-Computational-Structural-Biology 61 stars
- **Path-Planning** — Python package for P2: path planning for masked diffusion model sampling for sequence generation.
github.com/pengzhangzhi/Path-Planning 26 stars
- **Flash-Attention-with-Bias-Triton** — Triton implementation of Flash Attention with bias.
github.com/pengzhangzhi/Flash-Attention-with-Bias-Triton 25 stars

Honors and Awards

- Outstanding Paper Award, DeLTa Workshop at ICLR 2025
- NSF Travel Award, CIKM 2024
- Duke BME Travel Award, BMES 2024

Teaching

- Short Course Instructor. “Generative Models for Protein, Cell, and Biomedical Data”. ENAR 2026 Spring Meeting, with Anru R. Zhang and Alexander Tong. 3/2026

Reviewing

- Conference Reviewer: Neural Information Processing Systems (NeurIPS) 2025
- Conference Reviewer: International Conference on Learning Representations (ICLR) 2025
- Conference Reviewer: ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2025
- Workshop Reviewer: NeurIPS Workshop on Efficient Reasoning 2025
- Journal Reviewer: IEEE/ACM Transactions on Computational Biology and Bioinformatics
- Journal Reviewer: Computational and Structural Biotechnology Journal